

On Modal Logic of Full Products of Neighborhood Frames

Rajab Aghamov¹ Andrey Kudinov^{2,3} Maik Nguyen¹ Jakob Piribauer¹

¹Technische Universität Dresden

²Higher School of Modern Mathematics

³HSE University

June 4, 2025

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s
- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$ (com) and $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$ (chr) are valid

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s
- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$ (com) and $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$ (chr) are valid
- $S4 \times S4 = S4 \otimes S4 + (\text{com}) + (\text{chr})$

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s
- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$ (com) and $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$ (chr) are valid
- $S4 \times S4 = S4 \otimes S4 + (\text{com}) + (\text{chr})$
- (com) and (chr) fails on products of topological frames

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s
- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$ (com) and $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$ (chr) are valid
- $S4 \times S4 = S4 \otimes S4 + (\text{com}) + (\text{chr})$
- (com) and (chr) fails on products of topological frames
- $S4 \times_t S4 = S4 \otimes S4$

Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s
- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$ (com) and $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$ (chr) are valid
- $S4 \times S4 = S4 \otimes S4 + (\text{com}) + (\text{chr})$
- (com) and (chr) fails on products of topological frames
- $S4 \times_t S4 = S4 \otimes S4$
- $S4 \times_t^+ S4 = S4 \otimes S4 \otimes S4 + \Box p \rightarrow \Box_1p \wedge \Box_2p$

Contribution



- $S4 \times_t S4 = S4 \otimes S4$
- $S4 \times_t^+ S4 = S4 \otimes S4 \otimes S4 + \Box p \rightarrow \Box_1 p \wedge \Box_2 p$

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

Neighbourhood frame

Definition

Let X be a non-empty set and $k \geq 1$. A k -neighborhood frame (n-frame) is a pair (X, τ) , where $\tau : X \rightarrow 2^{2^X}$ assigns to every $x \in X$ a *filter* $\tau(x)$ on X , i.e.

- $\tau(x) \neq \emptyset$
- if $U \in \tau(x)$ and $U \subseteq V$ then $V \in \tau(x)$
- if $U, V \in \tau(x)$ then $U \cap V \in \tau(x)$

Full Product of Frames

Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathbf{X} \times \mathbf{Y}, \tau'_1, \tau'_2, \tau)$ defined as follows :

$$\tau'_1(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\},$$

$$\tau'_2(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\},$$

$$\tau(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) \exists W \in \tau_2(y) : V \times W \subseteq U\}$$

Full Product of Frames

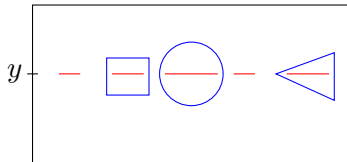
Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathbf{X} \times \mathbf{Y}, \tau'_1, \tau'_2, \tau)$ defined as follows :

$$\tau'_1(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\},$$

$$\tau'_2(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\},$$

$$\tau(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) \exists W \in \tau_2(y) : V \times W \subseteq U\}$$



Definition

Let L_1 and L_2 be two unimodal logics. Then the *fusion* ($\otimes$) of these logics is

$$K_2 + (L_1)_{\Box \rightarrow \Box_1} + (L_2)_{\Box \rightarrow \Box_2}$$

Furthermore the full product ($\times_n^+$) is defined as

$$L_1 \times_n^+ L_2 = \text{Log}\{\mathcal{X} \times \mathcal{Y} \mid \mathcal{X} \models L_1 \text{ and } \mathcal{Y} \models L_2\}$$

Main Theorem

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

In the following we will sketch the proof for $\Lambda = T$.

We denote $T \otimes T \otimes T + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$ as T_3m

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $Log(T_{\omega,\omega,\omega}^{r,m}) = T_3m$
- Note: For any Kripke frame $F = (W, R_1, \dots, R_n)$, we can construct a neighborhood frame F' , i.e. for any $w \in W$: $\tau_i(w) = \{U \mid R_i(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(F')$

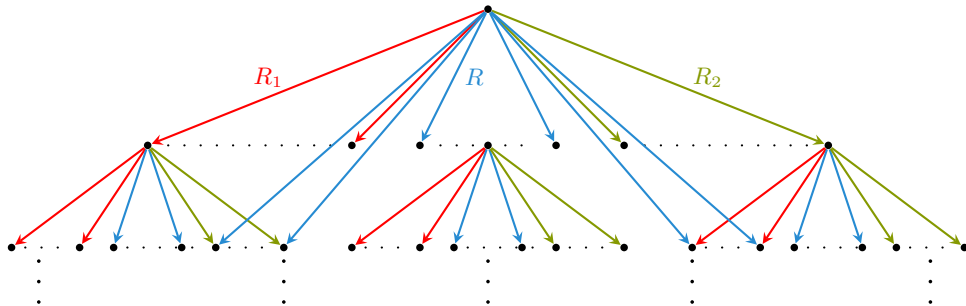
$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

Proof steps

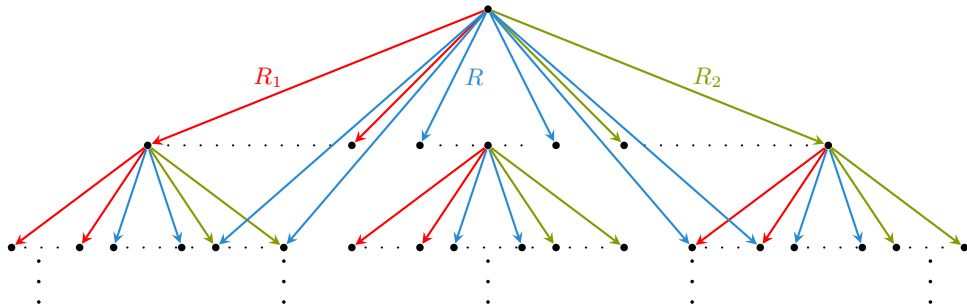
- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $Log(T_{\omega,\omega,\omega}^{r,m}) = T_3m$
- Note: For any Kripke frame $F = (W, R_1, \dots, R_n)$, we can construct a neighborhood frame F' , i.e. for any $w \in W$: $\tau_i(w) = \{U \mid R_i(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(F')$
- construct a neighbourhood product $N_{\omega}^r \times_n^+ N_{\omega}^r$ s.t. there is surjective bounded morphism from $N_{\omega}^r \times_n^+ N_{\omega}^r$ to $N(T_{\omega,\omega,\omega}^{r,m})$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

$$\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3 m$$



$$\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$$



- Note: $\Box p \rightarrow \Box_1 p \wedge \Box_2$ is derivable
- $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) \subseteq T_3m$ by standard techniques

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences.

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

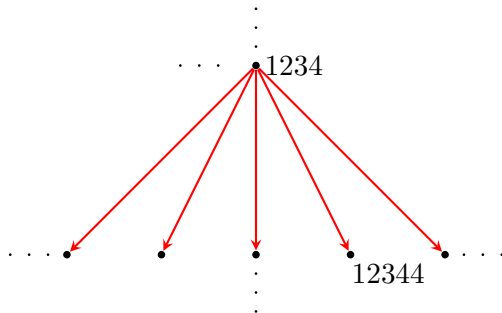
$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences. Let $\alpha = a_0a_1a_2... \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \text{forg}(\alpha)R\text{forg}(\beta) \text{ and } \alpha \upharpoonright_m = \beta \upharpoonright_m, \text{ with } m = \max(k, st(\alpha))\}$$

where $\text{forg}(\alpha)$ deletes all $\star$ and $ls(\alpha) = \min\{N \mid \forall k \geq N : a_k = \star\}$

Example of $U_k(\alpha)$



Let $\alpha = \star\star 12 \star 3 \star 4 \star\star^\omega \in X$. Then $ls(\alpha) = 9$. Let's consider $U_9(\alpha)$. Then

$$\star\star 12 \star 3 \star 4 \star 4 \star\star^\omega \in U_9(\alpha)$$

Definition of the neighborhood frame N_w^r

Definition

Let $T_\omega^r = (\mathbb{N}^*, R)$ be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2\ldots \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

is the set of pseudo infinite sequences. Let $\alpha = a_0a_1a_2\ldots \in X$. Then

$$U_k(\alpha) = \{\beta \in X \mid \text{forg}(\alpha)R\text{forg}(\beta) \text{ and } \alpha \upharpoonright_m = \beta \upharpoonright_m, \text{ with } m = \max(k, \text{st}(\alpha))\}$$

where $\text{forg}(\alpha)$ deletes all $\star$ and $\text{ls}(\alpha) = \min\{N \mid \forall k \geq N : a_k = \star\}$

We define $N_\omega^r = (X, \tau)$ where $\tau : X \rightarrow 2^{2^X}$ with

$$\tau(\alpha) = \{U \subseteq X \mid \exists k \in \mathbb{N} : U_k(\alpha) \subseteq U\}$$

Note: τ is a filter and there is no smallest neighborhood

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_w^r) = T$ and hence $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$*

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_w^r) = T$ and hence $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$*

Proof sketch: We fix a bijection $h : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$.

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_w^r) = T$ and hence $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$

Proof sketch: We fix a bijection $h : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Let $\alpha = a_0 a_1 \dots$ and $\beta = b_0 b_1 \dots \in X$
We define

$g : X \times X \rightarrow (\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*$ with $g(\alpha, \beta) = f \circ g(f(a_0 b_0) f(a_1 b_1) f(a_2 b_2) \dots)$

$$f(a, b) = \begin{cases} \star, & \text{if } a = b = \star, \\ (a, 1), & \text{if } b = \star, a \in \mathbb{N}, \\ (b, 2), & \text{if } a = \star, b \in \mathbb{N}, \\ (h(a, b), 0), & \text{otherwise.} \end{cases}$$

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_w^r) = T$ and hence $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$

- surjectivity and
- $R(g(\alpha, \beta)) \subseteq g(U_m(\alpha) \times U_m(\beta)) \subseteq g(U)$ are not hard to see

Note: $\tau(x, y) = \{U \mid \exists k, j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}, \sigma(x) = \{U \mid R(x) \subseteq U\}$.

2nd cond: for all $1 \leq i \leq k$, for all $w \in W$ and all $U \in \tau_i(w)$, there is a $V \in \sigma_i(f(w))$ with $V = f(U)$

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_w^r) = T$ and hence $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$

Pick $m > \max\{st(\alpha), st(\beta)\}$ and set $U := U_m(\alpha) \times U_m(\beta)$. Then

$$g(U) = g(U_m(\alpha) \times U_m(\beta)) \subseteq R(g(\alpha, \beta)) \subseteq V$$

Note: $\tau(x, y) = \{U \mid \exists k, j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}, \sigma(x) = \{U \mid R(x) \subseteq U\}$

3rd cond: for all $1 \leq i \leq k$, for all $w \in W$ and all $V \in \sigma_i(f(w))$, there exists $U \in \tau_i(w)$ such that $f(U) \subseteq V$

Surjective bounded morphism

Theorem

Let $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$ and $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$ be two full product frames. Then there is a surjective bounded morphism $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$. Note: $\text{Log}(N_\omega^r) = T$ and hence $N_\omega^r \times_n^+ N_\omega^r \in T \times_n^+ T$

Pick $m > \max\{st(\alpha), st(\beta)\}$ and set $U := U_m(\alpha) \times U_m(\beta)$. Then

$$g(U) = g(U_m(\alpha) \times U_m(\beta)) \subseteq R(g(\alpha, \beta)) \subseteq V$$

Remark: if $m \not> \max\{st(\alpha), st(\beta)\}$ then the subset relation fails. Example: Let $\alpha = 12\star^\omega, \beta = \star 23\star^\omega$ and $m = 1$. Then $g(\alpha, \beta) = (1, 1)h(2, 2)(3, 2)$. Assume $\alpha' = 123\star^\omega$. Hence, $g(\alpha', \beta) = (1, 1), h(2, 2)h(3, 3) \notin R(g(\alpha, \beta))$